# 44.4: The nephron is organized for stepwise processing of blood filtrate

In the human kidney, filtrate forms when fluid passes from the bloodstream to the lumen of Bowman's capsule. The glomerular capillaries and specialized cells of Bowman's capsule retain blood cells and large molecules, such as plasma proteins, but are permeable to water and small solutes. Thus, the filtrate produced in the capsule contains salts, glucose, amino acids, vitamins, nitrogenous wastes, and other small molecules. Because such molecules pass freely between glomerular capillaries and Bowman's capsule, the concentrations of these substances in the initial filtrate are the same as those in blood plasma.

Under normal conditions, roughly 1,600 L of blood flows through a pair of human kidneys each day, yielding about 180 L of initial filtrate. Both the volume and the composition of the filtrate are changed dramatically as processing occurs. About 99% of the water and nearly all of the sugars, amino acids, vitamins, and other organic nutrients are reabsorbed into the blood, leaving only about 1.5 L of urine to be transported to the bladder.

## From Blood Filtrate to Urine: *A Closer Look*

To explore how filtrate is processed into urine, we'll follow the filtrate along its path in the nephron. Each circled number in the text and the figure refers to the processing in transport epithelia as the filtrate moves through the kidney cortex and medulla **(Figure 44.13)**.

**Figure 44.13**

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    The nephron and collecting duct: regional functions of the
transport epithelium.
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Path of filtrate in the nephron and collecting duct outlined.

*The numbered regions in this diagram are keyed to the circled numbers in the text discussion of kidney function.*

Path of filtrate in the nephron and collecting duct outlined. Numbers in parentheses correspond to corresponding figure and text description numbers.

Overview: Nephron consists of proximal (1) and distal (4) tubules connected by U-shaped descending (2) and ascending (3) loop of Henle. The distal tubule connects to the collecting duct (5). Filtrate entering the

proximal tubule (1) consists of: H 2 0, salts (N a C l and others), H C O 3 superscript minus, H plus, urea, glucose, amino acids, some drugs.

Proximal tubule: Located in the cortex. H C O 3 superscript minus, H 2 O and K plus exit the proximal tubule and enter the interstitial fluid via passive transport. N a C l and nutrients exit the proximal tubule via active transport. H+ is actively transported from the interstitial fluid into the proximal tubule and NH3 enters the proximal tubule via passive transport.

Descending limb of loop of Henle: Spans the cortex, outer medulla and inner medulla. H2O exits the loop via passive transport in the outer medulla.

Thin segment of ascending limb: Located in the inner medulla. N a C l is passively transported out of the loop in the inner medulla.

Thick segment of ascending limb: Located in the outer medulla. NaCl is transported out of the loop actively in the outer medulla.

Distal tubule: Located in the cortex. N a C l and H C O 3 superscript minus are actively transported out of the tubule. H 2 O exits the tubule passively. K plus and H plus are actively transported into the tubule.

Collecting duct: Spans the cortex, outer and inner medulla. N a C l exits the duct via active transport in the outer medulla. Urea and H 2 O are passively transported out of the duct in the inner medulla.

*Some cells lining tubules in the kidney maintain normal cell volume by synthesizing organic solutes. Where in the kidney would you expect to find these cells? Explain.*

*For suggested answer, see*Appendix A*.*

**Animation**

Nephron Function

image

Play

image

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**1 Proximal tubule.** Reabsorption in the proximal tubule is critical for the recapture of ions, water, and valuable nutrients from the huge volume of initial filtrate. NaCl (salt) in the filtrate enters the cells of the transport epithelium by facilitated diffusion and cotransport mechanisms. There, Na+ ions are transferred to the interstitial fluid by active transport (see Concept 7.4). This transfer of positive charge out of the tubule drives the passive transport of Cl−.

As salt moves from the filtrate to the interstitial fluid, water follows by osmosis, reducing filtrate volume considerably. The salt and water that exit the filtrate diffuse from the interstitial fluid into the peritubular capillaries. Glucose, amino acids, potassium ions ($K^+$), and other essential substances are also actively or passively transported from the filtrate to the interstitial fluid and then into the peritubular capillaries.

Processing of filtrate in the proximal tubule helps maintain a relatively constant pH in body fluids. Cells of the transport epithelium secrete $H^+$ into the lumen of the tubule but also synthesize and secrete ammonia, which acts as a buffer to trap $H^+$ in the form of ammonium ions ($NH_4^+$). The more acidic the filtrate is, the more ammonia the cells produce and secrete, and a mammal's urine usually contains some ammonia from this source (even though most nitrogenous waste is excreted as urea). The proximal tubules also reabsorb about 90% of the buffer bicarbonate ($HCO_3^-$) from the filtrate, contributing further to pH balance in body fluids.

As the filtrate passes through the proximal tubule, materials to be excreted become concentrated. Many wastes leave the body fluids during the nonselective filtration process and remain in the filtrate while water and salts are reabsorbed. Urea, for example, is reabsorbed at a much lower rate than are salt and water. In addition, some materials are actively secreted into the filtrate from surrounding tissues. For example, drugs and toxins that have been processed in the liver pass from the peritubular capillaries into the interstitial fluid. These molecules are then actively secreted by the transport epithelium into the lumen of the proximal tubule.

**2 Descending limb of the loop of Henle.** Upon leaving the proximal tubule, filtrate enters the loop of Henle, which further reduces filtrate volume via distinct stages of water and salt movement. In the first portion of the loop, the descending limb, numerous water channels formed by **aquaporin** proteins make the transport epithelium freely permeable to water. In contrast, there are almost no channels for salt and other small solutes, resulting in very low permeability for these substances.

For water to move out of the tubule by osmosis, the interstitial fluid bathing the tubule must be hyperosmotic to the filtrate. This condition is met along the entire length of the descending limb because the osmolarity of the interstitial fluid increases progressively from the cortex through the medulla. As a result, the filtrate loses water and increases in solute concentration all along its journey down the descending limb. The highest osmolarity (about 1,200 mOsm/L in humans) occurs at the elbow of the loop of Henle.

**3 Ascending limb of the loop of Henle.** The filtrate reaches the tip of the loop and then returns to the cortex in the ascending limb. Unlike the descending limb, the ascending limb has a transport epithelium that lacks water channels. Consequently, the epithelial membrane that faces the filtrate in the ascending limb is impermeable to water.

The ascending limb has two specialized regions: a thin segment near the loop tip and a thick segment adjacent to the distal tubule. As filtrate ascends in the thin segment, NaCl, which became highly

concentrated in the descending limb, diffuses out of the permeable tubule into the interstitial fluid. This movement of NaCl out of the tubule helps maintain the osmolarity of the interstitial fluid in the medulla.

In the thick segment of the ascending limb, the movement of NaCl out of the filtrate continues. Here, however, the epithelium actively transports NaCl into the interstitial fluid. As a result of losing salt but not water, the filtrate becomes progressively more dilute as it moves up to the cortex in the ascending limb of the loop.

Although the loop of Henle has a small net effect on filtrate composition, it is a major site for the recovery of water (descending loop) and salt (ascending loop) from the filtrate. It is this recovery that underlies water conservation in land-dwelling vertebrates, as we will explore shortly.

**4 Distal tubule.** The distal tubule plays a key role in regulating the $K^+$ and NaCl concentration of body fluids. This regulation involves variation in the amount of $K^+$ secreted into the filtrate as well as the amount of NaCl reabsorbed from the filtrate. The distal tubule also contributes to pH regulation by the controlled secretion of $H^+$ and reabsorption of $HCO_3^-$.

**5 Collecting duct.** The collecting duct processes the filtrate into urine, which it carries to the renal pelvis (see Figure 44.12). As filtrate passes along the transport epithelium of the collecting duct, hormonal control of permeability and transport determines the extent to which the urine becomes concentrated.

When the kidneys are conserving water, aquaporin channels in the collecting duct allow water molecules to cross the epithelium. At the same time, the epithelium remains impermeable to salt and, in the renal cortex, to urea. As the collecting duct traverses the gradient of osmolarity in the kidney, the filtrate becomes increasingly concentrated, losing more and more water by osmosis to the hyperosmotic interstitial fluid. In the inner medulla, the duct becomes permeable to urea. Because of the high urea concentration in the filtrate at this point, some urea diffuses out of the duct and into the interstitial fluid. Along with NaCl, this urea contributes to the high osmolarity of the interstitial fluid in the medulla. The net result is urine that is hyperosmotic to the general body fluids.

When producing dilute rather than concentrated urine, the collecting duct actively absorbs salts without allowing water to follow by osmosis. At these times, the epithelium lacks aquaporin channels, and NaCl is actively transported out of filtrate. As we'll see, the presence of water channels in the collecting duct epithelium is controlled by hormones that regulate blood pressure, volume, and osmolarity.

## Solute Gradients and Water Conservation

The ability of the mammalian kidney to conserve water is a key adaptation for terrestrial habitats. In humans, the osmolarity of blood is about 300 mOsm/L, but the kidney can excrete urine up to four times as concentrated—about 1,200 mOsm/L. Some mammals can do even better: Australian hopping mice, small

marsupials that live in dry desert regions, can produce urine with an osmolarity of 9,300 mOsm/L, 25 times as concentrated as the animal's blood.

In a mammalian kidney, the production of hyperosmotic urine is possible only because considerable energy is expended for the active transport of solutes against concentration gradients. The nephrons—particularly the loops of Henle—can be thought of as energy-consuming machines that produce an osmolarity gradient suitable for extracting water from the filtrate in the collecting duct. The primary solutes affecting osmolarity are NaCl, which is concentrated in the renal medulla by the loop of Henle, and urea, which passes across the epithelium of the collecting duct in the inner medulla.

## Concentrating Urine in the Mammalian Kidney

To better understand the physiology of the mammalian kidney as a water-conserving organ, let's retrace the flow of filtrate through the excretory tubule. This time, let's focus on how juxtamedullary nephrons maintain an osmolarity gradient in the tissues that surround the loop of Henle and how they use that gradient to excrete hyperosmotic urine **(Figure 44.14)**. Filtrate passing from Bowman's capsule to the proximal tubule has about the same osmolarity as blood. A large amount of water *and* salt is reabsorbed from the filtrate as it flows through the proximal tubule in the renal cortex. As a result, the filtrate's volume decreases substantially, but its osmolarity remains about the same.

**Figure 44.14**

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How the human kidney concentrates urine.
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Overview of the osmolarity of the filtrate as it travels through the excretory tubule. Solutes are most concentrated near the elbow of loop of Henle, allowing salts to diffuse out.

Parts are described as follows.

Osmolarity of interstitial fluid: Cortex: 300; Outer medulla: 400 going to 600; Inner medulla: 900 going to 1200.

Proximal tubule (Cortex): Osmolarity in the tubule is 300; $H_2O$ moves out of tubule by passive transport near the renal outer medulla.

Descending limb of loop of Henle (Outer medulla): Osmolarity in nephron is 400 near the cortex going to 600 near the inner medulla. $H_2O$ flows out of tubule via passive transport.

Elbow of loop of Henle (Inner medulla): Osmolarity in tubule is 900 going to 1200. Water flows out passively.

Thin segment of ascending limb (Inner medulla): Osmolarity in tubule is 1200 going to 700. N a C l is passively transported out of the tubule and urea is passively transported into the tubule from the interstitial fluid.

Thick segment of ascending limb (Outer medulla): Osmolarity in the tubule is 400 going to 200. N a C l is actively transported out of the tubule.

Distal tubule (Cortex): Osmolarity in tubule is 100. N a C l is actively transported out of the tubule.

Collecting duct (spans cortex, outer and inner medulla): Osmolarity in the tubule is 300 in the cortex; 400 going to 600 in the outer medulla; and is 1200 in the inner medulla. H 2 O is passively transported out of the tubule in the cortex and the outer and inner medullas. N a C l is actively transported out of the tubule in the outer medulla. Urea is passively transported out of the tubule in the inner medulla.

Two solutes contribute to the osmolarity of the interstitial fluid: NaCl (used as shorthand here to refer collectively to Na+ and Cl−) and urea. The loop of Henle maintains the interstitial gradient of NaCl, which increases continuously in concentration from the cortex to the inner medulla. Urea diffuses into the interstitial fluid of the medulla from the collecting duct (although most of the urea in the filtrate remains in the collecting duct and is excreted). The filtrate makes three trips between the cortex and medulla: first down, then up, and then down again in the collecting duct. As the filtrate flows in the collecting duct past interstitial fluid of increasing osmolarity, more water moves out of the duct by osmosis. The loss of water concentrates the solutes, including urea, that will be excreted in the urine.

**WHAT IF?** *The drug furosemide blocks the cotransporters for* Na+ *and* Cl− *in the ascending limb of the loop of Henle. What effect would you expect this drug to have on urine volume?*

*For suggested answer, see* *Appendix A*.*

### Figure 44.14 Walkthrough

How the human kidney concentrates urine: the two-solute model

Watch an explanation of this figure in the Figure 44.14 Walkthrough: How the Human Kidney Concentrates Urine

image

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image

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As the filtrate flows from cortex to medulla in the descending limb of the loop of Henle, water leaves the tubule by osmosis. Solutes, including NaCl, become more concentrated, increasing the osmolarity of the filtrate. Diffusion of salt out of the tubule is maximal as the filtrate rounds the curve and enters the ascending limb, which is permeable to salt but not to water. NaCl diffusing from the ascending limb helps maintain a high osmolarity in the interstitial fluid of the renal medulla.

The loop of Henle and surrounding capillaries act as a type of countercurrent system to generate the steep osmotic gradient between the medulla and cortex. Recall that some endotherms have a countercurrent heat exchanger that reduces heat loss and that countercurrent gas exchange in fish gills maximizes oxygen absorption (see Figures 40.13 and 42.21). In those cases, the countercurrent mechanisms involve passive movement along either an oxygen concentration gradient or a heat gradient. In contrast, the countercurrent system of the loop of Henle involves active transport and thus an expenditure of energy. The active transport of NaCl from the filtrate in the upper part of the ascending limb of the loop maintains a high salt concentration in the interior of the kidney, enabling the kidney to form concentrated urine. Such countercurrent systems, which expend energy to create concentration gradients, are called **countercurrent multiplier systems**.

What prevents the capillaries of the **vasa recta** from dissipating the gradient by carrying away the high concentration of NaCl in the medulla's interstitial fluid? As shown in Figure 44.12, the descending and ascending vessels of the vasa recta carry blood in opposite directions through the kidney's osmolarity gradient. As the descending vessel conveys blood toward the inner medulla, water is lost from the blood and NaCl is gained by diffusion. These net fluxes are reversed as blood flows back toward the cortex in the ascending vessel of the vasa recta, with water reentering the blood and salt diffusing out. Thus, the vasa recta can supply the kidney with nutrients and other important substances carried by the blood without interfering with the osmolarity gradient in the inner and outer medulla.

The countercurrent-like characteristics of the loop of Henle and the vasa recta help to generate the steep osmotic gradient between the medulla and cortex. However, diffusion will eventually eliminate any osmotic gradient within animal tissue unless energy is expended to maintain the gradient. In the kidney, this expenditure largely occurs in the thick segment of the ascending limb of the loop of Henle, where NaCl is actively transported out of the tubule. Even with the benefits of countercurrent exchange, this process—along with other renal active transport systems—consumes considerable ATP. Thus, for its size, the kidney has one of the highest metabolic rates of any organ.

As a result of active transport of NaCl out of the thick segment of the ascending limb, the filtrate is actually hypoosmotic to body fluids by the time it reaches the distal tubule. Next, the filtrate descends again toward the medulla, this time in the collecting duct, which is permeable to water but not to salt. Therefore, osmosis extracts water from the filtrate as it passes from cortex to medulla and encounters interstitial fluid of

increasing osmolarity. This process concentrates salt, urea, and other solutes in the filtrate. Some urea passes out of the lower portion of the collecting duct and contributes to the high interstitial osmolarity of the inner medulla. (This urea is recycled by diffusion into the loop of Henle, but continual leakage from the collecting duct maintains a high interstitial urea concentration.) When the kidney concentrates urine maximally, the urine reaches 1,200 mOsm/L, the osmolarity of the interstitial fluid in the inner medulla. Although *isoosmotic* to the inner medulla's interstitial fluid, the urine is *hyperosmotic* to blood and interstitial fluid elsewhere in the body. This high osmolarity allows the solutes remaining in the urine to be excreted from the body with minimal water loss.

## Adaptations of the Vertebrate Kidney to Diverse Environments

**EVOLUTION** Vertebrates occupy habitats ranging from rain forests to deserts and from some of the saltiest bodies of water to the nearly pure waters of high mountain lakes. Comparing vertebrates across environments reveals adaptive variations in nephron structure and function. In the case of mammals, for example, the presence of **juxtamedullary nephrons** is a key adaptation that enables these terrestrial animals to shed salts and nitrogenous wastes without squandering water. Differences among species in the length of the loop of Henle in the juxtamedullary nephrons and in the relative numbers of juxtamedullary and **cortical nephrons** help to fine-tune osmoregulation to particular habitats.

### Mammals

Mammals that excrete the most hyperosmotic urine, such as Australian hopping mice, North American kangaroo rats, and other desert mammals, have many juxtamedullary nephrons with loops of Henle that extend deep into the medulla. Long loops maintain steep osmotic gradients in the kidney, resulting in urine becoming very concentrated as it passes from cortex to medulla in the collecting ducts.

In contrast, beavers, muskrats, and other aquatic mammals that spend much of their time in fresh water and rarely face problems of dehydration have mostly cortical nephrons, resulting in a much lower ability to concentrate urine. Terrestrial mammals living in moist conditions have loops of Henle of intermediate length and the capacity to produce urine intermediate in concentration to that produced by freshwater and desert mammals.

### Case Study: *Kidney Function in the Vampire Bat*

The South American vampire bat in **Figure 44.15** illustrates the versatility of the mammalian kidney. This species feeds at night on the blood of large birds and mammals. The bat uses its sharp teeth to make a small incision in the prey's skin and then laps up blood from the wound (the prey is typically not seriously harmed). Anticoagulants in the bat's saliva prevent the blood from clotting.

**Figure 44.15**

A vampire bat (*Desmodus rotundus*), a mammal with unique excretory challenges.

A bat.

A vampire bat may search for hours and fly long distances to locate a suitable victim. When it does find prey, it benefits from consuming as much blood as possible. Often drinking more than half its body mass, the bat is at risk of becoming too heavy to fly. As the bat feeds, however, its kidneys excrete large volumes of dilute urine, up to 24% of body mass per hour. Having lost enough weight to take off, the bat can fly back to its roost in a cave or hollow tree, where it spends the day.

In the roost, the vampire bat faces a different regulatory problem. Most of the nutrition it derives from blood comes in the form of protein. Digesting proteins generates large quantities of urea, but roosting bats lack access to the drinking water necessary to dilute it. Instead, their kidneys shift to producing small quantities of highly concentrated urine (up to 4,600 mOsm/L), an adjustment that disposes of the urea load while conserving as much water as possible. The vampire bat's ability to alternate rapidly between producing large amounts of dilute urine and small amounts of very hyperosmotic urine is an essential part of its adaptation to an unusual food source.

## Birds and Other Reptiles

Most birds, including the albatross (see Figure 44.1) and the ostrich **(Figure 44.16)**, live in environments that are dehydrating. Like mammals but no other species, birds have kidneys with juxtamedullary nephrons. However, the nephrons of birds have loops of Henle that extend less far into the medulla than those of mammals. Thus, bird kidneys cannot concentrate urine to the high osmolarities achieved by mammalian kidneys. Although birds can produce hyperosmotic urine, their main water conservation adaptation is having uric acid as the nitrogenous waste molecule.

**Figure 44.16**

An ostrich (*Struthio camelus*), an animal well adapted to its dry environment.

An ostrich.

The kidneys of other reptiles have only cortical nephrons, and they produce urine that is isoosmotic or hypoosmotic to body fluids. However, the epithelium of the cloaca from which urine and feces leave the body conserves fluid by reabsorbing water from these wastes. Like birds, most other reptiles excrete their nitrogenous wastes as uric acid.

## Freshwater Fishes and Amphibians

Hyperosmotic to their surroundings, freshwater fishes produce large volumes of very dilute urine. Their kidneys, which are packed with cortical nephrons, produce filtrate at a high rate. Salt conservation relies on the reabsorption of ions from the filtrate in the distal tubules.

Amphibian kidneys function much like those of freshwater fishes. When frogs are in fresh water, their kidneys excrete dilute urine while their skin accumulates certain salts from the water by active transport. On land, where dehydration is the most pressing problem of osmoregulation, frogs conserve body fluid by reabsorbing water across the epithelium of the urinary bladder.

## Marine Bony Fishes

Compared with freshwater fishes, marine fishes have fewer and smaller nephrons, and their nephrons lack a distal tubule. In addition, their kidneys have small glomeruli or lack glomeruli entirely. In keeping with these features, filtration rates are low and very little urine is excreted.

The main function of kidneys in marine bony fishes is to get rid of divalent ions (those with a charge of 2+ or 2−) such as calcium ($Ca^{2+}$), magnesium ($Mg^{2+}$), and sulfate ($SO_4^{2-}$). Marine fishes take in divalent ions by incessantly drinking seawater. They rid themselves of these ions by secreting them into the proximal tubules of the nephrons and excreting them in urine. Osmoregulation in marine bony fishes also relies on specialized *chloride cells* in the gills. By establishing ion gradients that enable secretion of salt (NaCl) into seawater, the chloride cells maintain proper levels of monovalent ions (charge of 1+ or 1−) such as $Na^+$ and $Cl^-$.

The generation of ion gradients and the movement of ions across membranes are central to salt and water balance in marine bony fishes. These events, however, are by no means unique to these organisms nor to homeostasis. As illustrated by the examples in **Figure 44.17**, osmoregulation by chloride cells is but one of many diverse physiological processes that are driven by the movement of ions across a membrane.

### Figure 44.17

**AP** Make Connections: Ion Movement and Gradients

Make connections: ion movement and gradients.

Transport of ions across the plasma membrane of a cell is a fundamental activity of all animals, and indeed of all living things. By generating ion gradients, ion transport provides the potential energy that powers processes ranging from an organism's regulation of salts and gases in internal fluids to its perception of and locomotion through its environment.

Osmoregulation: In marine bony fishes, ion gradients drive secretion of salt (N a C l), a process essential to avoid dehydration. Within gills, the pumps, cotransporters, and channels of specialized chloride cells

function together to drive salt from the blood across the gill epithelium and into the surrounding salt water. See figure 44.4.

Image shows that in a marine bony fish, N a plus is pumped out of the blood through a channel between chloride cells and into the saltwater environment as potassium is transported into a chloride cell from the blood. C l minus is pumped out of the chloride cell and into the saltwater. K plus is pumped from the chloride cell into the blood.

Gas Exchange: Ion gradients provide the basis for the opening of plant stomata by surrounding guard cells. Active transport of H plus out of a guard cell generates a voltage (membrane potential) that drives inward movement of K plus ions. This uptake of K plus by guard cells triggers an osmotic influx of water that changes cell shape, bowing the guard cells outward and thereby opening the stoma. See concept 36.4.

Information processing: In neurons, the opening and closing of channels selective for sodium or other ions underlies the transmission of information as nerve impulses. These signals enable nervous systems to receive and process input and to direct appropriate output, such as the leap of this frog capturing prey. See concept 48.3 and concept 50.5. Image: In the neuron of a frog leaping for prey, N a plus is transported across the membrane of the neuron through an open channel. When the channel is close, N a plus can not cross.

Locomotion: A gradient of H plus ions powers the bacterial flagellum. An electron transport chain generates this gradient, establishing a higher concentration of H plus outside the bacterial cell. Protons reentering the cell provide a force that causes the flagellar motor to rotate. The rotating motor turns the curved hook, causing the attached filament to propel the cell.

**BioFlix® Animation**

Membrane Transport

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